

YU-WEN CHEN

MACHINE LEARNING RESEARCHER - PHYSICAL AI & SATELLITE REMOTE SENSING

Ph.D. Candidate, Atmospheric & Oceanic Sciences | University of Colorado Boulder | NASA FINESST Graduate Fellow

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PROFESSIONAL SUMMARY

Ph.D. Candidate specializing in **Physical AI** and spatiotemporal numerical modeling. NASA FINESST Graduate Fellow with a track record of architecting PyTorch pipelines that integrate physics simulations (radiative transfer) with deep learning (Transformers/CNNs). Expert in solving inverse problems and scaling training on large-scale datasets (3M+ frames).

SKILLS

Programming: Python (expert - daily for data analysis, simulation, and tooling/wrappers), Fortran (radiative-transfer / atmospheric model codebases), Bash/Shell scripting

Machine Learning & Deep Learning: PyTorch, DeepXDE, scikit-learn, Physics-Informed Neural Networks (PINNs), CNNs, LSTMs, Transformers, Diffusion Models, Gradient-Boosted Trees (XGBoost/GBDT), Deep Ensembles, Tabular Deep Learning (TabM)

Scientific Computing & Data: NumPy, Pandas, Xarray, SciPy, Numba (JIT), Matplotlib, NetCDF/HDF5 (h5py/pyhdf), Time Series Analysis, Statistical Inference, Bayesian Optimization, PCA / Dimensionality Reduction, Quantile Regression, Conformal Prediction & Uncertainty Quantification, Cross-Validation & Hyperparameter Optimization

High-Performance & Cloud Computing: Linux, HPC, SLURM (CU Research Computing), Git version control, AWS

Physical Modeling & Remote Sensing: 3D Radiative Transfer (EaR3T), Spatiotemporal Numerical Modeling, Satellite Remote Sensing, Inverse Problems, Multi-Sensor Satellite Collocation (OCO-2/MODIS), Geospatial Geometry (ECEF/WGS84, KD-Tree nearest-neighbor search)

AI-Augmented Development: Agentic AI tools (Claude, GitHub Copilot) across the research-to-production lifecycle - literature synthesis, experimental design, prototype-to-production scaling, automated pipeline engineering

Domain Expertise: Atmospheric Physics, Carbon Monitoring (OCO-2 / CO₂), Air Quality & Pollutant Transport, Chemistry

PROFESSIONAL EXPERIENCE

University of Colorado Boulder

Jan. 2022 - Present

NASA FINESST Graduate Fellow / Graduate Researcher

- NASA FINESST fellowship runs Dec. 2024 - Nov. 2026 (2 years).
- Used EaR3T 3D radiative transfer modeling to understand the physical mechanism of cloud-induced CO₂ retrieval biases and develop a physics-based correction method, achieving **76% error reduction** in cloud-adjacent scenes - published in *Atmos. Meas. Tech.* (2025).
- Advancing bias correction through spectral analysis: isolating 3D cloud-effect signatures encoded in OCO-2 spectra, then predicting and removing XCO₂ retrieval anomalies at scale with an ensemble of supervised models - gradient-boosted trees (XGBoost), deep ensembles with Gaussian-NLL heads, and tabular deep learning (TabM) - calibrated with conformal prediction for reliable uncertainty intervals and validated via date-blocked k-fold cross-validation.
- Built the OCO-2/MODIS collocation pipeline underlying this work: matching glint-mode OCO-2 footprints to Aqua MODIS cloud masks and computing per-sounding nearest-cloud distance via ECEF conversion and banded KD-Tree search, with Numba-accelerated processing on the CU Research Computing (SLURM) cluster.
- Applied agentic AI tools (Claude, GitHub Copilot) as both a **scientific research accelerator** (literature synthesis, hypothesis exploration, experimental design, results interpretation) and an **engineering accelerator** (scaled EaR3T prototype to production OCO-2 processing, automated end-to-end data workflows, optimized algorithm speed and memory, generated test and validation suites) - compressing the research-to-deployment cycle.

- Managed collaborations across 4 research institutes and presented findings at 11 oral and 11 poster presentations, including venues in the Netherlands, Japan, Greece, and Taiwan.

University of Colorado Boulder

Aug. 2024 - Present

Graduate Researcher - Arctic Surface Properties

- Engineered an automated intelligent system using Python to autonomously detect and correct sensor anomalies.
- Functioned as a **self-healing data agent** for high-fidelity Arctic radiometric measurements.
- Implemented instrument normalization protocols to standardize raw outputs from diverse radiometric sensors.

Academia Sinica, Research Center for Environmental Changes (RCEC)

Jan. 2019 - Jul. 2021

Research Assistant - Pollution Transport Analysis

- Developed autoregressive generative models (CNN/LSTM) to predict station-level ozone evolution in complex terrain.
- Investigated variable importance in ground-level ozone formation with supervised learning, leading to an oral presentation at **AI: Modeling Oceans and Climate Change (AIMOCC 2021) Workshop at ICLR 2021**.

EDUCATION

Ph.D., Atmospheric and Oceanic Sciences - University of Colorado Boulder (NASA FINESST Fellow) Aug. 2021 - Dec. 2026 (expected)

M.S., Atmospheric and Oceanic Sciences - University of Colorado Boulder Aug. 2021 - May 2023

M.S., Chemistry - National Taiwan University Aug. 2016 - Jun. 2018

B.S., Chemistry (Minor: Atmospheric Science) - National Taiwan University Aug. 2013 - Jun. 2016

PUBLICATIONS

PUBLISHED

1. **Yu-Wen Chen**, K. Sebastian Schmidt, Hong Chen, Steven T. Massie, Susan S. Kulawik, and Hironobu Iwabuchi (2025). Mitigation of satellite OCO-2 CO₂ biases in the vicinity of clouds with 3D calculations using the Education and Research 3D Radiative Transfer Toolbox (EaR3T). *Atmos. Meas. Tech.*, 18, 1859-1884. doi:10.5194/amt-18-1859-2025
2. **Yu-Wen Chen**, Sourav Medya, Yi-Chun Chen (2022). Investigating variable importance in ground-level ozone formation with supervised learning. *Atmos. Environ.*, 119148. doi:10.1016/j.atmosenv.2022.119148
3. Min-Ken Hsieh, **Yu-Wen Chen**, Yi-Chun Chen, Chien-Ming Wu (2022). The Roles of Local Circulation and Boundary Layer Development in Tracer Transport over Complex Topography in Central Taiwan. *J. Meteorol. Soc. Japan*, 100, 555-573.
4. **Yu-Wen Chen**, Arun P. Periasamy, Huan-Tsung Chang, Chien-Fu Chen (2019). Quantification of glucose via in situ growth of Cu₂O/Ag nanoparticles. *Sens. Actuator B-Chem.*, 285, 224-231.
5. Arun P. Periasamy, Pavithra Sriram, **Yu-Wen Chen**, Chien-Wei Wu, Ta-Jen Yen, Huan-Tsung Chang (2019). Porous aluminum electrodes with 3D channels and zig-zag edges for efficient hydrogen evolution. *Chem. Commun.*, 55, 5447-5450.
6. Rini Ravindranath, Arun P. Periasamy, Prathik Roy, **Yu-Wen Chen**, Huan-Tsung Chang (2018). Smart app-based on-field colorimetric quantification of mercury via analyte-induced enhancement of the photocatalytic activity of TiO₂-Au nanospheres. *Anal. Bioanal. Chem.*, 410, 4555-4564.
7. Rini Ravindranath, Prathik Roy, Arun P. Periasamy, **Yu-Wen Chen**, Chi-Te Liang, Huan-Tsung Chang (2017). Fe₂O₃/Al₂O₃ microboxes for efficient removal of heavy metal ions. *New J. Chem.*, 41, 7751-7757.

ACCEPTED (IN FINAL REVISION)

1. Sebastian Becker, Konrad Sebastian Schmidt, Hong Chen, **Yu-Wen Chen**, Kerry G. Meyer, Colten A. Peterson, and Manfred Wendisch (2026). Bias in satellite-derived cloud radiative effect over Arctic sea ice relative to aircraft measurements during ARCSIX. *Atmos. Chem. Phys.* (accepted 2026-08-27, in final revision)

IN REVIEW

1. Patrick C. Taylor, Armin Sooroshian, Rei Ueyama, Sebastian Schmidt, ... **Yu-Wen Chen**, ... and Paquita Zuidema (2026). The Arctic Radiation-Cloud-Aerosol-Surface Interaction Experiment (ARCSIX) airborne campaign dataset. *Earth Syst. Sci. Data Discuss.*

IN PREPARATION

1. **Yu-Wen Chen**, Ken Hirata, Hong Chen, Vikas Nataraja, and K. Sebastian Schmidt (2026). High Arctic Sea Ice Albedo Shifts Cloud Radiative Effect Toward Surface Warming. *Geophys. Res. Lett.*
2. **Yu-Wen Chen** et al. (2026). Correcting cloud-proximity biases in OCO-2 XCO₂ with photon path-length statistics and deep ensembles, validated against TCCON, aircraft, and shipborne observations. *Atmos. Meas. Tech.*

PRESENTATIONS

ORAL PRESENTATIONS (* PRESENTING AUTHOR)

1. **Diagnosing and Correcting Cloud-Proximity Biases in OCO-2 XCO₂ with Photon Path-Length Statistics**
Yu-Wen Chen* | SKATERS (Scientists for Knowledge of ATmospheric invERSe modeling) student-organized seminar, remote | August 2026
2. **Spring- and Summertime Airborne Observations in the Arctic Suggest Prolonged Cloud-Induced Surface Melt**
Yu-Wen Chen*, Sebastian Schmidt, Ken Hirata, Vikas Nataraja, and Hong Chen | 2026 AMS Madison Summit, Madison, WI, US | August 2026
3. **Introduction and Demonstration of the Education and Research 3D Radiative Transfer Toolbox (EaR3T)**
Yu-Wen Chen*, Sebastian Schmidt, Hong Chen, Ken Hirata, and Vikas Nataraja | Netherlands Institute for Space Research (SRON), Leiden, Netherlands | May 2026
4. **Mitigation of Cloud-induced Biases in CO₂ Remote Sensing**
Yu-Wen Chen*, Sebastian Schmidt, Hong Chen, and Steven T. Massie | Netherlands Institute for Space Research (SRON), Leiden, Netherlands | May 2026
5. **Spring- and Summertime Airborne Observations in the Arctic Suggest Prolonged Cloud-Induced Surface Melt**
Yu-Wen Chen, Ken Hirata*, Sebastian Schmidt, Hong Chen, and Vikas Nataraja | ARCSIX Science Team Meeting 2026, Hampton, VA, US | April 2026
6. **Spring- and Summertime Airborne Observations in the Arctic Suggest Prolonged Cloud-Induced Surface Melt**
Yu-Wen Chen*, Sebastian Schmidt, Hong Chen, Ken Hirata, and Vikas Nataraja | University of Colorado Boulder, CO, US | April 2026
7. **Correcting Cloud-induced Biases in CO₂ Remote Sensing**
Yu-Wen Chen*, Sebastian Schmidt, Hong Chen, and Steven T. Massie | National Taiwan University, Taipei, Taiwan | January 2026
8. **The Impact of Polarization on OCO-2 Bias Mitigation in the Vicinity of Clouds**
Yu-Wen Chen*, Sebastian Schmidt, Steven T. Massie, and Susan S. Kulawik | 4th Advancement of POLarimetric Observations (APOLO), Kyoto, Japan | November 2024
9. **Satellite 3D Radiance Simulator for the OCO-2 Mission and its Application to the Mitigation of Spectroscopy Retrieval Biases in the Vicinity of Clouds**
Yu-Wen Chen*, Sebastian Schmidt, Steven T. Massie, and Susan S. Kulawik | International Radiation Symposium 2022, Thessaloniki, Greece | July 2022
10. **Investigating Ground-level Ozone Formation: A Case Study in Taiwan**
Yu-Wen Chen*, Sourav Medya, and Yi-Chun Chen | AI: Modeling Oceans and Climate Change (AIMOCC 2021) Workshop at ICLR 2021, virtual | May 2021
11. **Quantitation of glucose through its manipulation of the growth of Cu₂O/Ag nanoparticles**
Yu-Wen Chen*, Arun P. Periasamy, and Huan-Tsung Chang | Analytical Chemistry Technology Conference, Keelung, Taiwan | May 2018

POSTER PRESENTATIONS (* PRESENTING AUTHOR)

1. **Investigating 3D Cloud Radiative Effects on High-Resolution Trace Gas Spectra to Elucidate the Mechanisms of Land-Ocean Retrieval Biases**
Yu-Wen Chen*, Sebastian Schmidt, Hong Chen, and Steven T. Massie | 2026 AMS Madison Summit, Madison, WI, US | August 2026

2. **The Impact of 3D Cloud Radiative Effect on Trace Gas Retrievals: Bridging the Gap from Low Earth Orbit to Geostationary Missions**
Yu-Wen Chen*, Sebastian Schmidt, Hong Chen, and Steven T. Massie | EGU General Assembly 2026, Vienna, Austria | May 2026
3. **Cloud-Induced Warming of Spring and Summer Arctic Sea Ice is Likely Underestimated by Passive Satellite Imagery Derived Products**
Yu-Wen Chen*, Sebastian Schmidt, Hong Chen, Ken Hirata, Vikas Nataraja, and Sebastian Becker | Radiation and Climate Gordon Research Conference 2025, Lewiston, ME, US | July 2025
4. **The Impact of Polarization on OCO-2 Bias Mitigation in the Vicinity of Clouds**
Yu-Wen Chen*, Sebastian Schmidt, Steven T. Massie, and Susan S. Kulawik | 4th Advancement of POLarimetric Observations, Kyoto, Japan | November 2024
5. **Mitigation of OCO-2 CO₂ Biases in the Vicinity of Cloud**
Yu-Wen Chen*, Sebastian Schmidt, Steven T. Massie, and Susan S. Kulawik | 20th International Workshop on Greenhouse Gas Measurements from Space, Boulder, CO, US | May 2024
6. **Mitigating the Impact of Three-Dimensional Cloud Effects on CO₂ Retrievals for Various Surface and Cloud Types Using a 3D Radiance Simulator**
Yu-Wen Chen*, Sebastian Schmidt, Steven T. Massie, and Susan S. Kulawik | AGU Fall Meeting 2023, San Francisco, CA, US | December 2023
7. **Mitigation of Satellite OCO-2 CO₂ Biases in the Vicinity of Clouds with 3D Calculations**
Yu-Wen Chen*, Sebastian Schmidt, Steven T. Massie, and Susan S. Kulawik | OCO-2/3 Science Team Meeting, Boulder, CO, US | October 2023
8. **The Impact of Clouds on the Orbiting Carbon Observatory-2 Satellite Trace Gas Retrievals**
Yu-Wen Chen*, Sebastian Schmidt, Steven T. Massie, and Susan S. Kulawik | Workshop Lille 2023, Lille, France | May 2023
9. **Mitigation of OCO-2 Spectroscopy Retrieval Biases in the Vicinity of Clouds**
Yu-Wen Chen*, Sebastian Schmidt, Steven T. Massie, and Susan S. Kulawik | OCO-2/3 Science Team Meeting, Boulder, CO, US | October 2022
10. **Intercomparison of aircraft and ground based measurements of pollutants and aerosols near major pollution sources over Taiwan**
Yu-Wen Chen*, Yi-Chun Chen, Charles C.-K. Chou, et al. | 18th Asian Chemical Congress, Taipei, Taiwan | December 2018
11. **Quantitation of Glucose through its Manipulation of the growth of Cu₂O/Ag Nanoparticles**
Yu-Wen Chen*, Arun P. Periasamy, and Huan-Tsung Chang | Graduate Student Poster Session, NTU Department of Chemistry, Taipei, Taiwan | June 2018

TEACHING EXPERIENCE

Weather and the Atmosphere

Fall 2021

Teaching Assistant - University of Colorado Boulder

- Assisted in general affairs for 350 undergraduate students.
- Led weekly lecture review and exam study guides.

High School Student Research

Sept. 2016 - Jul. 2018

Research Instructor - National Taiwan University

- Instructed three high school students on fuel cell research; students received Advanced Honor in the Young Scientists Development Program.

Analytical Chemistry Lecture

Spring 2017

Teaching Assistant - National Taiwan University

- Assisted in general affairs for 160 undergraduate students.
- Led weekly lecture review and discussion for study groups.

Analytical Chemistry Lab

Fall 2016, Spring 2017

Teaching Assistant - National Taiwan University

- Led weekly experiment review and guided laboratory experiments.

CERTIFICATIONS

- **ECMWF MOOC Course 1: Foundations and New Frontiers** (Certificate of Completion) - ECMWF, 2026
- **ECMWF MOOC Course 2: Concepts of Machine Learning in Weather and Climate** (Certificate of Completion) - ECMWF, 2026

AWARDS & HONORS

- **NASA FINESST Award** (13% selection rate) - University of Colorado Boulder, 2024
- **Summer School for Inverse Modeling of Greenhouse Gases** - Colorado State University, 2024
- **Government Scholarship to Study Abroad** - Taiwan Ministry of Education, 2023
- **NASA Workshop Lille Travel Support** - University of Oklahoma, 2023
- **Best Popularity Award of Poster** - NTU Graduate Student Poster Session, 2018
- **Teaching Excellence Award** - National Taiwan University, 2017
- **NTU-Kanagawa University Chemistry Workshop** - Kanagawa University, Japan, 2014